

Oles Matsyshyn, Ph.D.

Quantitative Research | Time-Series and Risk Modeling | Machine Learning
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EDUCATION

TU DRESDEN

PHD IN PHYSICS

2018--2022 | Germany

Max Planck Institute for the Physics of Complex Systems
Magna cum laude

TARAS SHEVCHENKO UNIV.

MS IN PHYSICS

2016--2018 | Kyiv, Ukraine

Quantum Field Theory
Diploma with distinction

TARAS SHEVCHENKO UNIV.

BS IN PHYSICS

2012--2016 | Kyiv, Ukraine

Quantum Field Theory

SKILLS

PROGRAMMING

Python, MATLAB, C++, SQL
Git, NumPy, SciPy, pandas, scikit-learn

QUANTITATIVE METHODS

Hidden Markov models
Time-series analysis
Monte Carlo simulation
Optimization, backtesting
Risk and regime modeling

STATISTICS AND ML

GARCH / APARCH
ARIMA / EWMA
VaR / CVaR
XGBoost, PCA
Regression, classification
Survival analysis

SYSTEMS

IBKR API
HPC and parallel computing
Automated reporting

CERTIFICATIONS

MITx: Quant Finance
Wharton: Quant Modeling
Python for Finance

LANGUAGES

English: professional
Ukrainian: native
German: intermediate

PROFILE

Ph.D. physicist and quantitative researcher with 10+ years of experience translating mathematical models into validated computational systems. Combines statistical modeling, machine learning, time-series analysis, numerical optimization, and research software development in Python and MATLAB.

EXPERIENCE

INDEPENDENT QUANTITATIVE RESEARCH | PYTHON, IBKR API

2025--Present | Singapore

- Designed and backtested systematic equity and leveraged-ETF strategies using volatility, trend, regime, and risk-control signals.
- Developed and historically validated a 2-state hidden Markov regime model on daily US equity data from 1999–2026; high-risk states were followed by 2.2–2.4x higher 5-day and 2.1–2.2x higher 21-day realized volatility.
- Regime probabilities correlated 0.44–0.50 with forward 5-day volatility and 0.47–0.58 with forward 21-day volatility across validation specifications.
- Engineered an IBKR execution and monitoring platform with automated reconnection, scheduled trading, order tracking, persistent state, restart recovery, risk controls, and reporting.

NANYANG TECHNOLOGICAL UNIVERSITY | RESEARCH FELLOW

2022--Present | Singapore

- Led end-to-end computational research from mathematical formulation and numerical implementation to validation and publication, contributing to 12+ papers since 2022.
- Built Python/MATLAB pipelines for high-dimensional dynamical systems, stochastic evolution, optimization, and large parameter studies.
- Made multi-day simulations tractable using analytical model reduction, adaptive grids, parallel computing, convergence tests, and independent limiting-case checks.
- Contribute across 4 active international collaborations in Germany, China, and the United States; author of 17 published or posted papers overall, including 6 Physical Review Letters, with 5 projects in preparation.

SELECTED PROJECTS

CLINICAL SURVIVAL MODELING | PYTHON, ML

- Built an end-to-end survival-prediction pipeline for the QRT Data Challenge using blood-test and protein-mutation data from approximately 2,000 patients.
- Achieved a held-out C-index of 0.74 versus a 0.64 benchmark through feature engineering, imputation, validation, and tree-based modeling.

HIGH-DIMENSIONAL NUMERICAL MODELING | PYTHON, MATLAB, HPC

- Developed and validated multiband Hamiltonian and quantum-dynamics simulations for graphene systems, a 4D optimization framework for electron-phonon scattering in GaAs, and numerical models of superconducting quantum devices and qubits.
- Validated numerical results against analytical limits and stability, sensitivity, and convergence tests.